

ORB Animation Authoring Spec

Brief for artists creating animations for the Imagix Crystal Ball POV display

PERSISTENCE-OF-VISION · 90 LEDs · 8 COLORS · ~25 FPS

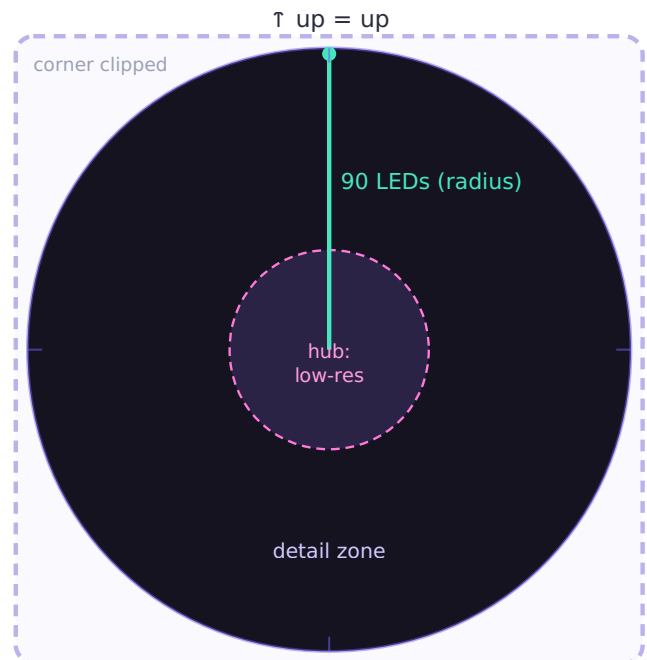
This device is a **persistence-of-vision (POV) holographic fan**: a single blade of **90 RGB LEDs** spins fast, painting a circular image one radial line at a time. One full revolution paints one frame. The specs below are reverse-engineered hardware limits — treat them as constraints, not suggestions, and the art will survive conversion intact.

Mental model: you are drawing **glowing neon on a black spinning disc**. Only lit pixels show; black is invisible (the LED is simply off).

1 Canvas & resolution

Spec	Value
Frame shape	Square , 1:1
Deliver at	1024×1024 px ideal 512×512 px minimum
Safe area	Centered circle only — corners are not displayed
Background	Pure black <code>#000000</code> (= off = invisible)
Effective res	~ 90 px radial (center→rim) ~ 565 px angular at the rim

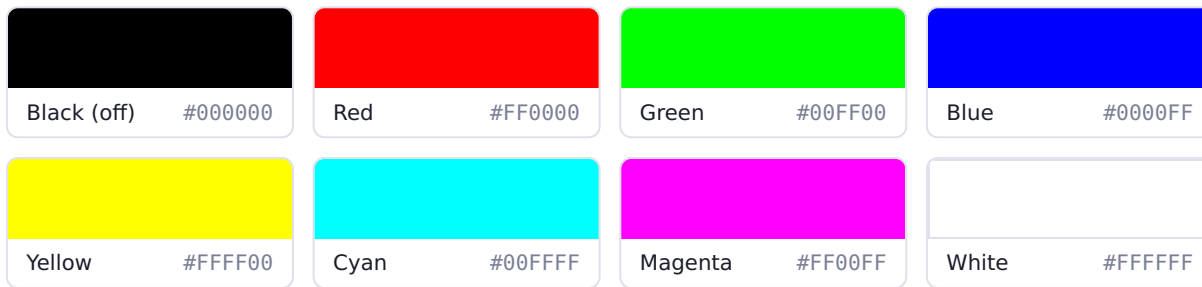
Author high; the encoder samples down. Detail finer than the effective grid is lost — **design bold, not fussy**. Strokes 2–3 px minimum (1 px lines flicker). Draw “up = up”; the fixed mounting offset is corrected at encode time, so **do not pre-rotate**.



The square you deliver → the circular disc it becomes. Keep detail in the outer ring; corners and hub are weak.

2 Color — the hard constraint

The display is **1 bit per channel**. That means **8 possible colors, full stop**. Everything else is approximated by **dithering** (a grainy stipple of these 8) — gradients, soft shading and anti-aliased edges all become visible noise.



For the crispest result: flat fills in these 8 colors, hard edges, high contrast. Mid-tones and pastels read poorly. Dithering can be a deliberate retro look — just author *knowing* it happens rather than being surprised by it.

3 Motion & frame counts

~25 frames per second — one painted frame per revolution (~1500 RPM).

Frame count = duration (seconds) × 25. A 4-second loop = ~100 frames.

Loops must be seamless: the last frame flows back into the first with no jump (idle, listening, thinking, talk, responding). Fast motion strobes at 25 fps — favor smooth, continuous movement.

Reference (measured)	Frames	Time
Factory power-on	116	~4.6 s
Factory boot logo	122	~4.9 s

Delivery: zero-padded PNG sequence preferred — [eb_idle_0001.png ... eb_idle_0100.png](#) (lossless, exact frames). Animated GIF acceptable as fallback.

4 State hierarchy — what to make, how many, how long

The assistant has a set of states. **Each numbered variant is a separate animation the device picks from at RANDOM** when that state fires — so to theme a state cleanly you must fill every variant, or leftovers from another set bleed in. You may also add *more* variants than listed; the count is not fixed. Frame counts are **recommended targets** (the director's call), not hard requirements.

Tier 1 — conversational loop (must-haves)

State	Variants	Length (frames / sec)	Type	Notes
idle	4	50-100 / 2-4 s	loop	Resting — the most-seen state. Make these distinct
wake	1	25-40 / 1-1.5 s	one-shot	Wake word — a "perk up"
vad	1	15-25 / ~1 s	short	Voice detected; quick acknowledgement
listening	4	50-75 / 2-3 s	loop	While the user is speaking
thinking	2	50-75 / 2-3 s	loop	Processing
meaning	1	~25 / ~1 s	one-shot	Parsing what was said
responding	7	50-100 / 2-4 s	loop	Plays while talking — most screen time, most variety
talk	1	25-50 / 1-2 s	loop	Mouth-move / speaking loop
confirm	4	25-50 / 1-2 s	one-shot	Acknowledgement

Tier 2 — personality / polish

State	Variants	Length (frames / sec)	Type	Notes
happy	1	40-75 / 1.5-3 s	one-shot	Positive reaction
error	1	40-60 / ~2 s	one-shot	Something went wrong
low	1	~40 / ~1.5 s	one-shot	Low battery
bye	1	50-75 / 2-3 s	one-shot	Goodbye
selected	1	25-40 / 1-1.5 s	one-shot	When chosen
start	1	25-50 / 1-2 s	one-shot	Startup
pwon	3	75-116 / 3-4.6 s	one-shot	Power-on intro
pwoff	3	50-75 / 2-3 s	one-shot	Power-off outro

System set (shared `bu_` — optional, lowest priority)

State	Variants	Length	Type	Notes
bootup	1	100-125 / 4-5 s	one-shot	Boot splash / logo
on	1	25-50 / 1-2 s	one-shot	Power-on
connecting	1	~50 / ~2 s	loop	Network bring-up (loops until connected)
connected	1	~25 / ~1 s	one-shot	Connection success
inprogress / updone	1 ea	25-50	mixed	Update progress / done

5 Delivery checklist

- ☐ Square frames, **1024×1024** (512² min), all content inside the centered circle
- ☐ Pure black background; design as glowing shapes (black = off = invisible)
- ☐ Palette limited toward the **8 displayable colors**; flat fills, hard edges, high contrast
- ☐ **Seamless** first→last frame on every looping state
- ☐ Important detail in the mid-to-outer radius (the hub is low-res)
- ☐ “Up = up” orientation; no pre-rotation
- ☐ PNG sequences, zero-padded, named by state + variant (`eb_responding_03_0001.png ...`)
- ☐ Frame counts per the tables; every variant of a themed state filled

6 Technical reference (pipeline-side — artists can ignore)

Polar geometry: **2100 columns / revolution × 90 LEDs / column**; 34 bytes/column (2 sync bits + 270 data bits = 90 × RGB, 1 bit/channel, MSB-first). One frame per revolution; ~25 rev/s ⇒ ~25 fps. Frame = 2100 × 34 = 71,400 B (~1.79 MB/s). The encoder `orb_encode.py` samples the square source into the 2100 × 90 polar grid, dithers to 1-bit RGB, and applies the fixed orientation offset via `--cpr 2100 --rotate <deg>`. Author resolution only needs to exceed the effective grid (90 radial / ~565 angular); 1024² gives clean anti-aliased downsampling headroom.